
Where Does Platonic Convergence Live? Pointwise Curvature of a Galaxy Embedding Manifold and What a Linear Map Can See

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations on galaxy images
3 using Mean-KNN (MKNN) and Centered Kernel Analysis (CKA). We ask whether
4 that convergence is concentrated where the embedding manifold is flat. A point-
5 wise curvature field, obtained by differentiating a trained decoder and validated
6 against analytic truth in that regime, finds mean-curvature magnitude correlating
7 negatively with cross-survey neighbour agreement at every latent dimension, with
8 local density explaining 50 to 80 percent of it. A second-order expansion of a
9 linear map restricted to the manifold explains why: a readout sees the second fun-
10 damental form contracted with its normal component, $\langle w_N, \Pi \rangle$, and an alignment
11 sees the relative second fundamental form of the two embeddings, neither visible
12 to mean-curvature magnitude at large codimension. On a unit-normalized em-
13 bedding that Hessian splits into a sphere term, proportional to the readout’s own
14 value, and a shape term. On a synthetic surface at production scale the sphere term
15 reads -0.74 to -0.80 across samplings that swing the $\|H\|$ partial from -0.29 to
16 $+0.26$; on 86,471 labelled galaxies it is negative on every label, the mismatch be-
17 tween the label’s own Hessian and the shape term is negative wherever the label
18 has local structure, and the shape term alone takes the label-dependent sign that
19 mismatch predicts. Convergence tests must condition on relative curvature, not on
20 one model’s.

21 1 Introduction

22 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different data
23 converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on the sky:
24 galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC) and embedded by
25 a family of different vision models share 0.4 to 2 percent of their nearest neighbours. We ask where
26 on the manifold that agreement lives: if two models disagree most where the embedding surface
27 bends sharply, local mean curvature should associate negatively with local cross-survey agreement.
28 We test this claim using a pointwise mean-curvature estimation from a trained autoencoder on the
29 embedding space.

30 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
31 trained decoder, validated against analytic truth in the production regime; (ii) a density-controlled
32 test of the curvature–alignment prediction under two probes, which finds no effect free of density;
33 (iii) the residual expansion of a linear map on a submanifold, which names the curvature such a map
34 responds to and shows mean-curvature magnitude is blind to it; and (iv) a known-surface and a real-

data test on the linear-readout side, where the expansion’s terms behave as predicted and account for a label-dependent sign.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019], from the Platonic Universe release [Duraphe et al., 2025], from which we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey ViT-B embeddings with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the read-out experiment. Embeddings are unit-normalized (data on S^{767}). We then estimate the manifold’s first homology ($\beta_1 = 0$ under bootstrap bands [Fasy et al., 2014]), thus we one Euclidean chart is appropriate for the autoencoder.

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; curvature is evaluated on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector is $H = \text{tr}_g \Pi$, by automatic differentiation checked against finite differences. On the unit sphere every d -submanifold carries a radial component of H equal to $-d$ that says nothing about shape. To account for this, the map differentiated is $F/\|F\|$ (after which $H_{\text{rad}} = -d$ to 10^{-14}) and $\|H_{\text{tan}}\|$ carries the shape. Decoder curvature has served as a regularizer [Lee and Park, 2023] and been read from templates [Acosta et al., 2022]; here it is an instrument, validated against analytic fixtures.

Validation. Scored on surfaces with closed-form curvature on four axes: rank (Spearman against true $\|H\|$), direction (cosine of H vectors), magnitude ratio and calibration. At $D = 768$ an in-sphere surface $G(z) = Q \text{ normalize}([\phi(z); 0.8 h(z); 0 \dots])$ (ϕ the inverse stereographic map $\mathbb{R}^{16} \rightarrow S^{16}$, h four Gaussian bumps, Q a fixed rotation, span 18 dimensions) at $n = 86,471$, $k = 2,048$ gives rank 0.94, direction cosine 0.999, magnitude ratio 1.00, and 0.75 rank under noise at 25% of patch radius.

3 Curvature and cross-survey convergence

Probe. Per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings, the Platonic Universe statistic made local, correlated with $\|H\|$ over all 10,000 points at $d \in \{20, 25, 32\}$. Centered kernel alignment [Kornblith et al., 2019] is the second probe.

Result. The raw association $\rho(\|H\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009), in the predicted direction. But k -NN statistics are density-sensitive [Radovanović et al., 2010]: density correlates with agreement at -0.21 at every d and with $\|H\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 : 78%, 49% and 8% of the association removed. Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; replicating across three decoder seeds; the CKA tertile gap within strata agrees at $d = 20$ and 25 , -0.05 to -0.08 against a null band of ± 0.017). Replacing $\|H\|$ by $\|H_{\text{tan}}\|$ collapses the $d = 25$ partial to -0.023 , into the range of the two d that fail. Convergence is not detectably concentrated by mean-curvature magnitude once density is controlled.

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^d$ be d -dimensional; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr } \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(|u|^3)$, so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

To second order, a linear function is as non-linear on M as M bends in that function’s normal direction [Lee, 2018]. For $M \subset S^{D-1}$ the normal bundle splits into the sphere’s radial direction and the

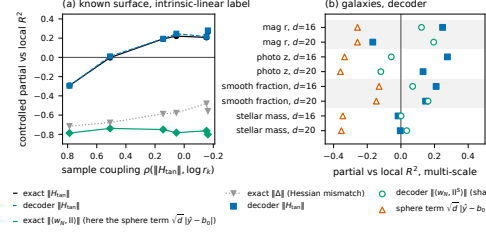


Figure 1: (a) Known surface, intrinsically linear label, exact geometry: three-control partial against local R^2 as the sample’s density–curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$: partial under the multi-scale density control for the decoder’s $\|H_{\tan}\|$ (filled squares), the shape term (open circles) and the sphere term (open triangles).

rest, $\Pi = \Pi^S - g \hat{x}$, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x}) g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates (the $\frac{1}{2}$ below uses Gaussian fourth moments), with $\bar{c}$ the patch-mean residual, $\delta = \nabla_M y - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2} s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2} [J_G(\Gamma^G - \Gamma^F) + \Pi_G - A \Pi_F](u, u) + O(|u|^3)$, with Γ the induced connections: the second-order obstruction to linear alignment is a connection mismatch plus the *relative* second fundamental form, and reduces to the latter when the induced metrics agree; its normal part is $\Pi_G - P_N^G A \Pi_F$.

So the curvature a linear map pays for is $\langle w_N, \Pi \rangle$ or $\Pi_G - P_N^G A \Pi_F$; $\|H\|$ bounds only a trace and need not correlate with either at codimension 750; the term carries s^4 , so patch radius, hence density, multiplies it; and when the label has intrinsic curvature of its own, Δ carries the residual and the curvature piece can take either sign.

5 The probe-facing curvature, on a known surface and on galaxies

The readout side can be tested directly: w is known and Π comes from the decoder. Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at $\alpha = 100$ from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial Spearman against local R^2 (Freedman–Lane permutation p -values [Freedman and Lane, 1983]) controlling for log patch radius, local label variance and count and, as a stronger control, log radius at $k \in \{16, 64, 256, 1024, 2048\}$. Per anchor we compute $\|H_{\tan}\|$ and the probe-facing Hessian $\|\langle w_N, \Pi \rangle\|_g$ with $w_N = w - Jg^{-1}J^\top w$, split by Eq. (2) into its shape and sphere terms.

Known surface. The validation surface is resampled with weight $\|H_{\tan}\|^\gamma$, so only where the points sit changes (Figure 1a). The $\|H_{\tan}\|$ partial follows the sampling from -0.29 at coupling $+0.80$ through zero to a plateau near $+0.22$, and the decoder reproduces it. The probe-facing Hessian $\|\langle w_N, \Pi \rangle\|$ reads -0.74 to -0.80 at every sampling. On this surface it is the sphere term of Eq. (2): rank 0.996–1.000 with $\sqrt{d}|\hat{y} - b_0|$, the in-sphere part 20 to 40 times smaller (one in-sphere normal direction) and, where resolvable, following $\|H_{\tan}\|$. The fixture thus confirms the expansion’s sphere term: an intrinsically linear label loses local R^2 where the readout’s own value is large, whatever the sampling. With a nonlinear label the sphere term reads null and $\|\Delta\|$ carries the partial (-0.2 to -0.6), as Eq. (3) predicts.

Galaxies. Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing Hessian (rank 0.96–0.99 with the full norm). To read it through Eq. (3) we estimate $\text{Hess}_M y$ per anchor by a least-squares quadratic of the label on the tangent-projected coordinates $u = g^{-1}J^\top(x - x_0)$ of its 2,048 neighbours, where the quadratic coefficient is the covariant Hessian. Three regularities (Table 1). The sphere term is negative on every label and both d (-0.13 to

label	$\ H_{\text{tan}}\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	−0.17	+0.12	+0.19	−0.26	−0.26	−0.39	−0.40	+0.35	+0.43
photo_z	+0.28	+0.13	−0.06*	−0.12	−0.33	−0.36	−0.45	−0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	−0.13	−0.15	−0.09*	−0.17	+0.19	+0.07*
stellar_mass	−0.02*	−0.00*	+0.00*	+0.03*	−0.35	−0.35	−0.04*	−0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control; alignment is $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$. * not significant at 0.05; the three-control partials agree in sign in every significant cell.

−0.36), as on the fixture. Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass is null throughout) the mismatch is negative (−0.09 to −0.45) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive (+0.07 to +0.43): local R^2 is higher where the manifold bends in the readout’s direction the way the label bends. The shape term alone is therefore label-signed (positive for magnitude and smooth fraction, null to weakly negative for redshift) because it is one of three terms in $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, \langle w_N, \Pi \rangle \rangle + \|\langle w_N, \Pi \rangle\|^2$: its sign follows the cross term’s where that is nonzero and the remaining square where it is null. $\|H_{\text{tan}}\|$ changes sign between $d = 16$ and 20 for magnitude: it sits on the curve of Figure 1a (coupling to radius −0.60). The same regression applied to the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median metric cosine with the decoder’s 0.55–0.65), and an independent split-half quadratic-chart estimator of Π gives the same shape-term sign on three of four labels.

Across encoders. The quadratic-chart estimator’s mean-bending trace, run on five encoders at the same 512 anchors ($d = 16$, $k = 2,048$: ViT-B, DINOv3, CLIP-B, ConvNeXt-B, ViT-L), gives density-controlled associations with the magnitude probe’s local R^2 of −0.24, 0.00, +0.27, +0.23 and −0.19. A trace statistic has no encoder-stable sign, as Eq. (1) predicts for a quantity the readout does not see; the terms of Eq. (3) are what a cross-encoder test should carry.

6 Discussion

Three things are established. A trained decoder yields a pointwise second fundamental form at production scale whose readout-facing component agrees with an independent estimator and with the point cloud itself. The residual expansion names what a linear map responds to, and on galaxies its terms behave as predicted: the sphere term and the Hessian mismatch carry the predicted negative sign, and the shape term is label-signed because it enters through its alignment with the label’s own Hessian. The same expansion explains why mean-curvature statistics disagree across estimators and encoders: they are traces of a tensor that each map contracts differently. For the Platonic question this fixes the design. MKNN and CKA are invariant to rigid motions of either embedding, so a per-point convergence score can be conditioned on curvature only through a relative quantity such as $\Pi_G - P_N^G A \Pi_F$, which needs both embeddings’ second fundamental forms and a map between tangent frames. Our pilot at $d = 20$ (2,048 anchors) supplies both and locates the obstruction to linear alignment at first order: a global ridge map carries HSC tangent frames into the Legacy embedding with 44% residual, and the second-order term shows no density-free association yet (+0.04, n.s.; −0.18 to −0.24 with a locally fitted tangent map, +0.13 decoder-free). Aligning tangent frames first and then testing relative curvature is the next step, and the instrument for it is in hand.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10–0.17; a 136-coefficient fit of the same kind failed a split-half reliability gate). The sphere term is not separated from ridge shrinkage of extreme predictions. Permutation p -values assume anchor exchangeability under overlapping patches.

References

Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. The platonic representation hypothesis. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL <https://arxiv.org/abs/2405.07987>.

157 Kshitij Duraphe, Michael J. Smith, Shashwat Sourav, and John F. Wu. The platonic universe: Do foundation
158 models see the same sky? In *NeurIPS 2025 Workshop on Machine Learning and the Physical Sciences*, 2025.
159 URL <https://arxiv.org/abs/2509.19453>.

160 Oriane Siméoni, Huy V. Vo, Maximilian Seitzer, Federico Baldassarre, Maxime Oquab, et al. DINOv3. *arXiv*
161 *preprint*, 2025. URL <https://arxiv.org/abs/2508.10104>.

162 Hiroaki Aihara, Nobuo Arimoto, Robert Armstrong, et al. The Hyper Suprime-Cam SSP Survey: Overview
163 and survey design. *Publications of the Astronomical Society of Japan*, 70(SP1):S4, 2018. doi: 10.1093/pasj/
164 psx066.

165 Arjun Dey, David J. Schlegel, Dustin Lang, et al. Overview of the DESI Legacy Imaging Surveys. *The Astro-*
166 *nomical Journal*, 157(5):168, 2019. doi: 10.3847/1538-3881/ab089d.

167 Michael J. Smith, Ryan J. Roberts, Eirini Angeloudi, and Marc Huertas-Company. AstroPT: Scaling large obser-
168 vation models for astronomy. *arXiv preprint*, 2024. URL <https://arxiv.org/abs/2405.14930>.

169 Brittany Terese Fasy, Fabrizio Lecci, Alessandro Rinaldo, Larry Wasserman, Sivaraman Balakrishnan, and Aarti
170 Singh. Confidence sets for persistence diagrams. *Annals of Statistics*, 42(6):2301–2339, 2014.

171 Yonghyeon Lee and Frank Chongwoo Park. On explicit curvature regularization in deep generative models. In
172 *2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG-ML)*, ICML, 2023.
173 URL <https://arxiv.org/abs/2309.10237>.

174 Francisco Acosta, Sophia Sanborn, Khanh Dao Duc, Manu Madhav, and Nina Miolane. Quantifying extrinsic
175 curvature in neural manifolds. *arXiv preprint*, 2022. URL <https://arxiv.org/abs/2212.10414>.

176 Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network rep-
177 resentations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*,
178 2019.

179 Miloš Radovanović, Alexandros Nanopoulos, and Mirjana Ivanović. Hubs in space: Popular nearest neighbors
180 in high-dimensional data. *Journal of Machine Learning Research*, 11:2487–2531, 2010.

181 John M. Lee. *Introduction to Riemannian Manifolds*, volume 176 of *Graduate Texts in Mathematics*. Springer,
182 2nd edition, 2018. doi: 10.1007/978-3-319-91755-9.

183 Guillaume Alain and Yoshua Bengio. Understanding intermediate layers using linear classifier probes, 2016.
184 arXiv:1610.01644.

185 David Freedman and David Lane. A nonstochastic interpretation of reported significance levels. *Journal of*
186 *Business & Economic Statistics*, 1(4):292–298, 1983. doi: 10.1080/07350015.1983.10509354.